

5 Solve the equation $3 \cos A - \sec A = 5 \tan A$ for $0 \leq A \leq 2\pi$.

[5]

$$3 \cos A - \frac{1}{\cos A} = \frac{5 \sin A}{\cos A} \quad (M1)$$

$$\frac{3 \cos^2 A - 1}{\cos A} = \frac{5 \sin A}{\cos A}$$

$$\frac{3 [1 - \sin^2 A] - 1}{\cos A} = \frac{5 \sin A}{\cos A} \quad (M1)$$

$$\frac{3 - 3 \sin^2 A - 1}{\cos A} = \frac{5 \sin A}{\cos A}$$

$$\frac{2 - 3 \sin^2 A}{\cos A} = \frac{5 \sin A}{\cos A}$$

$$\cos A [2 - 3 \sin^2 A - 5 \sin A] = 0$$

$$\div (-1), \quad \cos A [3 \sin^2 A + 5 \sin A - 2] = 0$$

$$\cos A (3 \sin A - 1)(\sin A + 1) = 0$$

$$\cos A = 0 \quad \text{or} \quad (3 \sin A - 1)(\sin A + 2) = 0 \quad (M1)$$

(rejected $\therefore$

$\frac{1}{\cos A}$ cannot be defined)

(A1)

$$\sin A = \frac{1}{3} > 0 \quad \text{or} \quad \sin A = -2 \quad (\text{rejected})$$

$$\text{ref } A = 0.33983 \dots$$

$$\therefore A = 0.33983, 2.801755 \dots$$

$$= 0.340, 2.80 \quad (3 \text{ sf})$$

(A1)